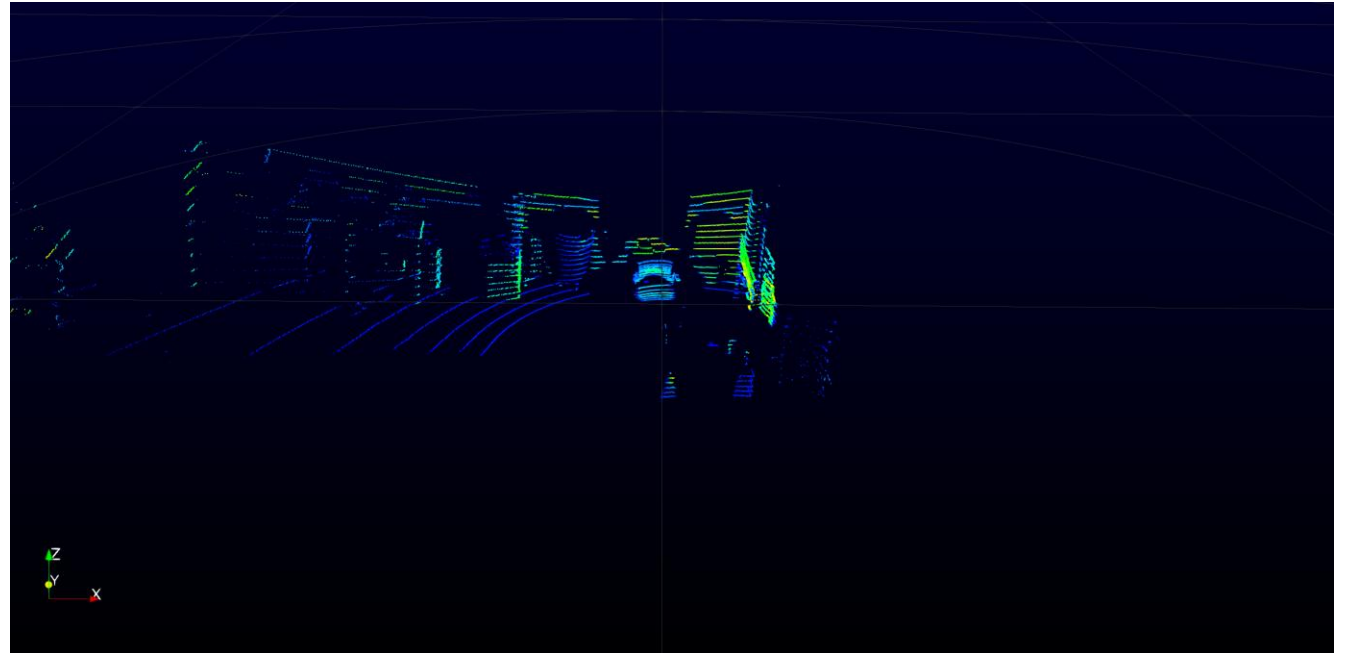


First Results

Estimated Relative Pose

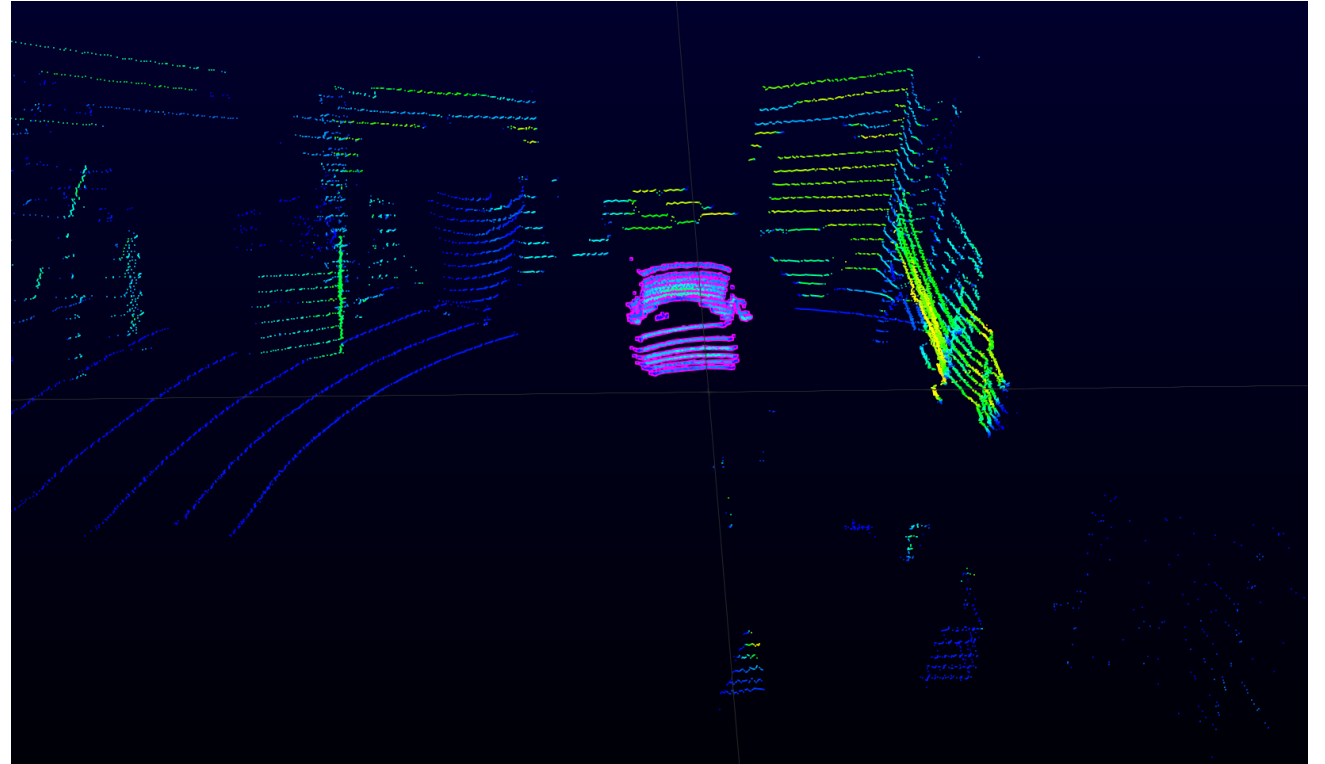
With the D-SVD method under arbitrary: transformation, roll, pitch, yaw relative to origin

VLP 16 LIDAR sensor creates a 360 degree Horizontal rendering, but with a 30-40 degree field of view using 16 lasers, limiting some of the transformations (pitch)



Origin

Tested origin frame versus translated frame



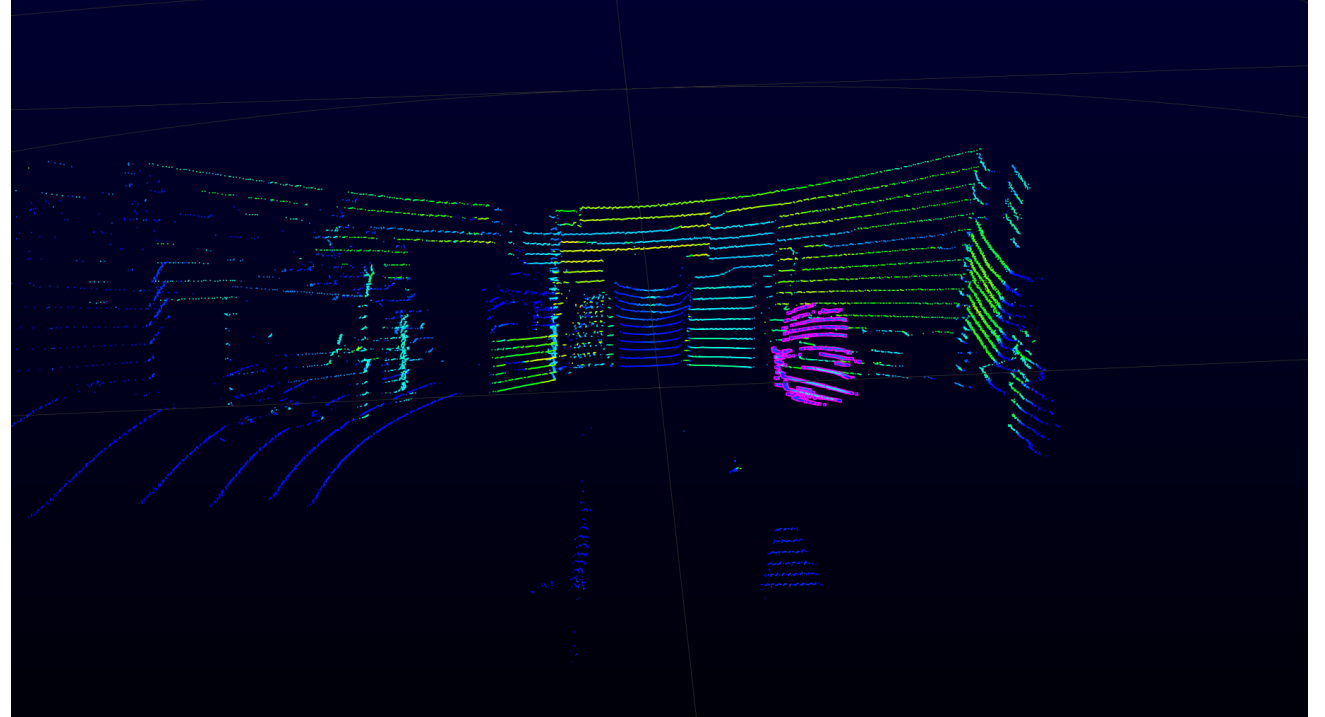
Translation

Estimated Rotation:

0.9136	0.3906	0.1127
-0.3730	0.9157	-0.1496
-0.1617	0.0946	0.9823

Estimated Translation:

0.8432
0.0006
-0.2146



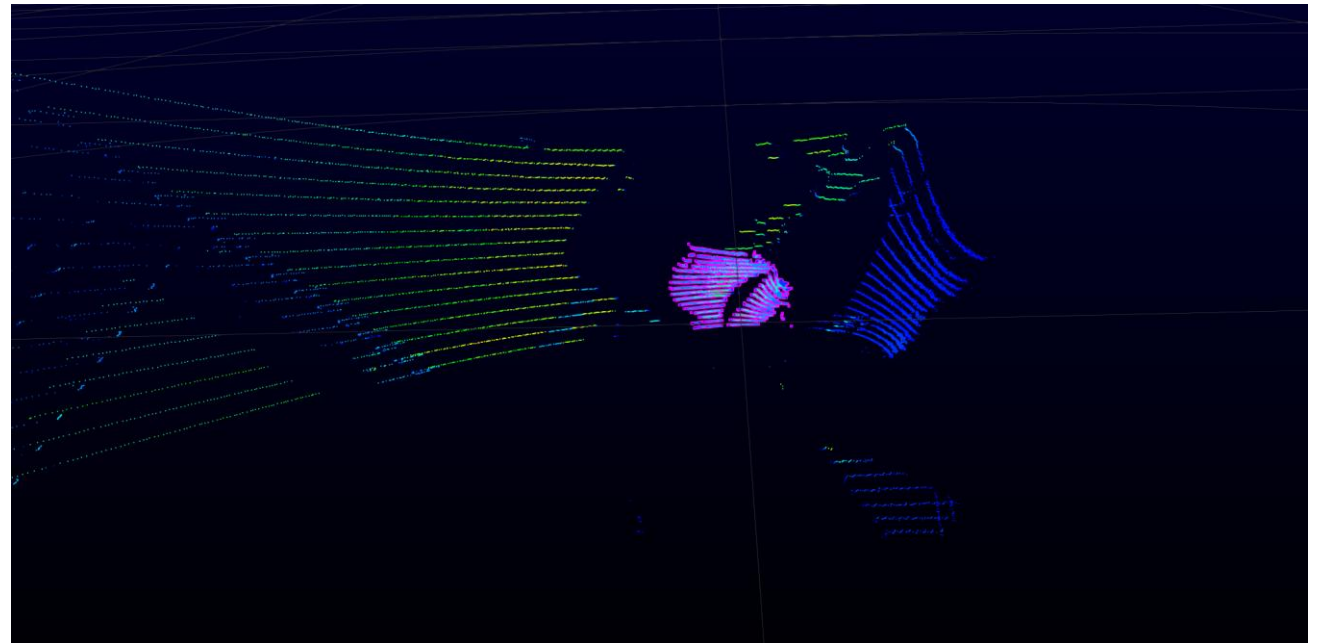
Roll

Estimated Rotation:

-0.2059	0.9472	0.2457
-0.4464	0.1325	-0.8850
-0.8708	-0.2918	0.3956

Estimated Translation:

-0.7460
0.5896
0.1395



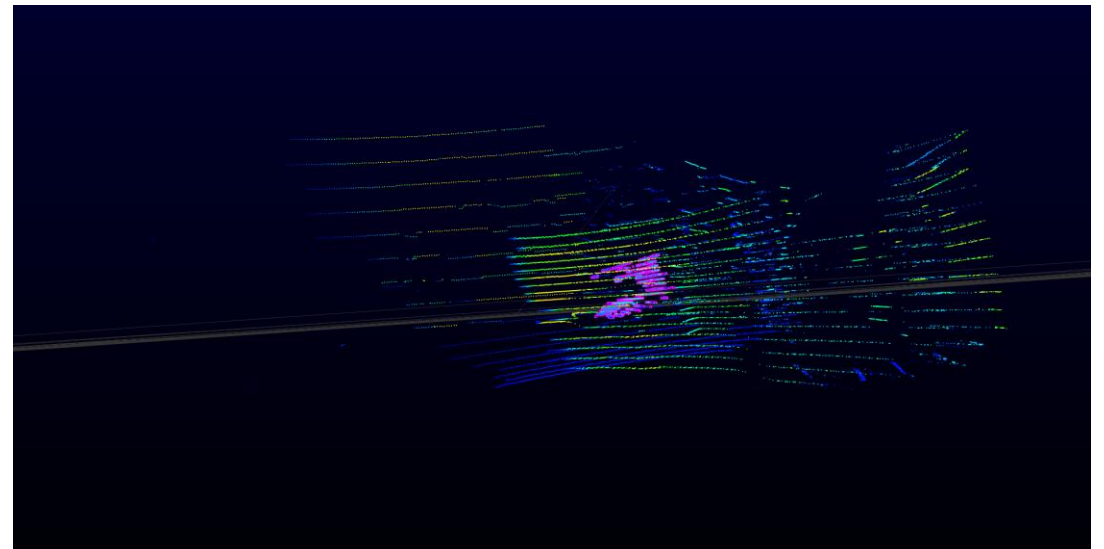
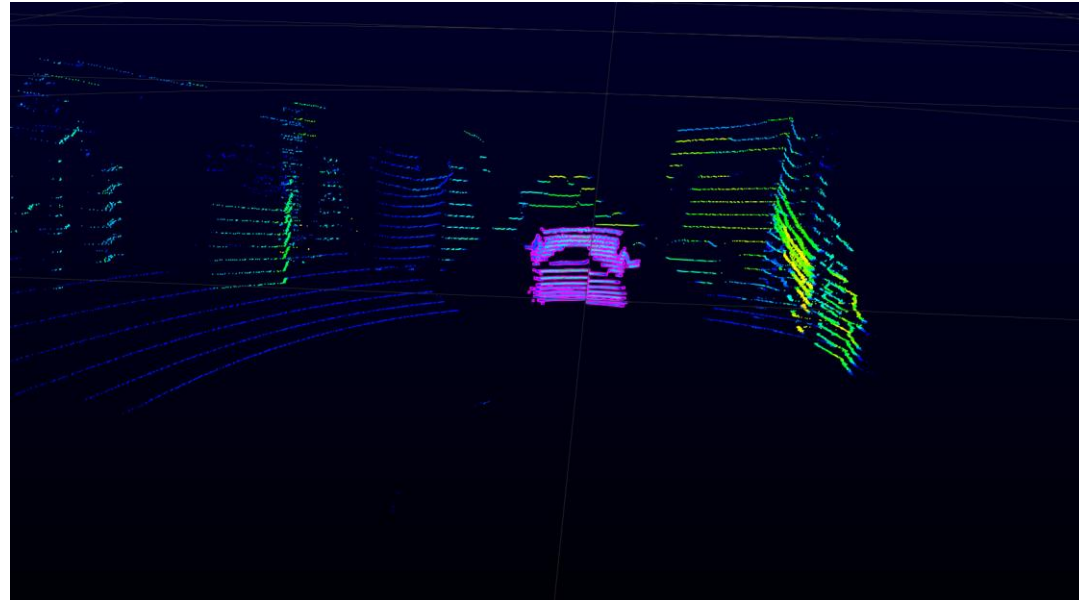
Pitch

Estimated Rotation:

0.9699	0.2400	0.0423
-0.2392	0.9707	-0.0215
-0.0463	0.0107	0.9989

Estimated Translation:

-0.1360
-0.0672
-0.0067



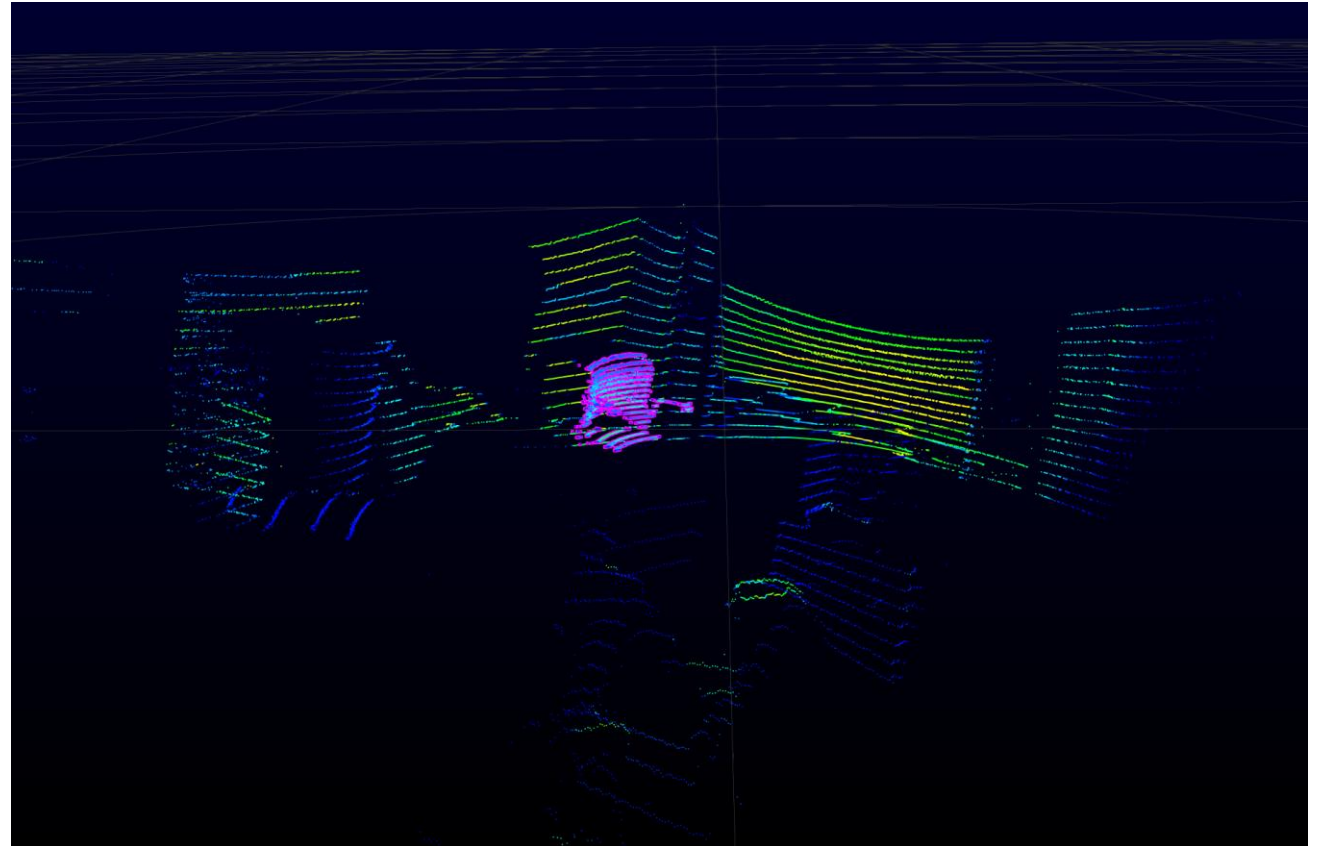
Yaw

Estimated Rotation:

0.5635	-0.1188	0.8175
0.0585	0.9929	0.1040
-0.8240	-0.0107	0.5665

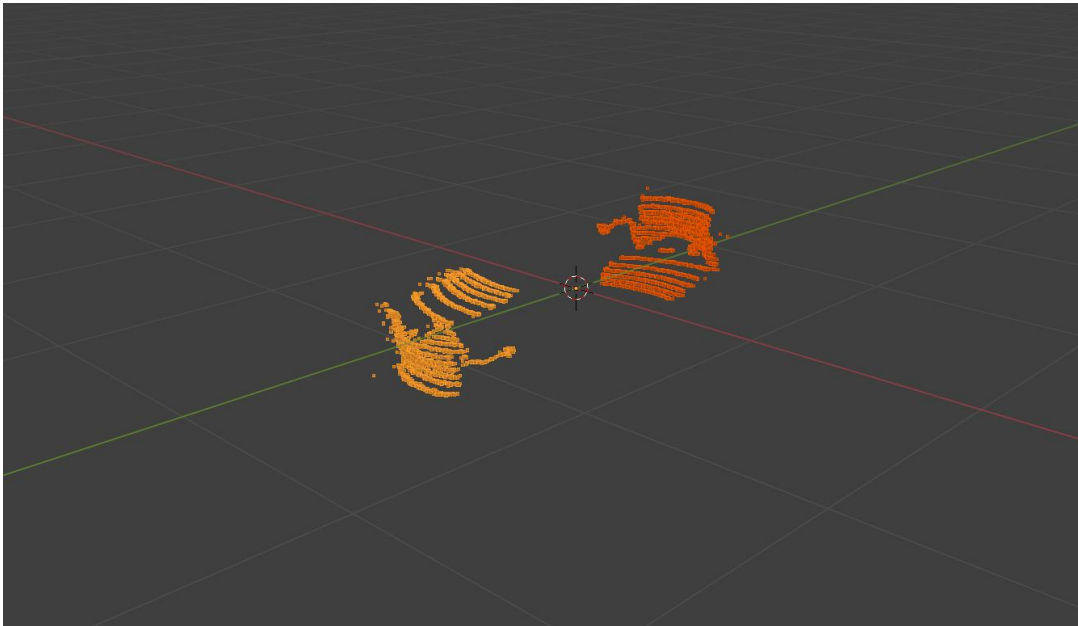
Estimated Translation:

-0.4319
-0.1851
-0.0638



Same number of points

Origin Isolated, Transformed (same number of points)



Results:

Estimated Rotation:

0.9577	0.0037	-0.2878
-0.0037	1.0000	0.0006
0.2878	0.0006	0.9577

Estimated Translation:

-0.0040
-1.4639
0.0078